

Team Project Proposal

Business Question

Can Airbnb use deep learning to identify high-demand NYC listings from listing title text, location, room type, price, host activity, availability, and map-derived spatial context?

Rationale: This could help Airbnb identify which listing features are associated with stronger guest activity, supporting host guidance, pricing advice, search ranking, and neighbourhood-level marketplace planning.

Fits Track 2 because we can use:

- a neural network for structured tabular features
- an NLP/text branch for the name column
- a CNN branch using image patches from the NYC map

Recommended target variable: demand

Since the dataset does not contain actual bookings, “demand” has to be a proxy. We can treat review activity as a proxy for demand.

High demand = 1 if reviews_per_month is in the top 25%

High demand = 0 otherwise

Notes:

- reviews_per_month has many missing values because listings with no reviews have no review rate. EDA task - prefill with zero.
- Inspect and exclude extreme values (outliers) before modelling.
- Do not use reviews_per_month and number_of_reviews as input features because they are closely related to review activity / demand, and could make the model look stronger than it really is.

Input to avoid (no value):

- id
- host_name
- host_id

Inputs:

- name
- neighbourhood_group
- neighbourhood
- latitude
- longitude
- room_type
- price
- minimum_nights
- calculated_host_listings_count
- availability_365

Model performance should be evaluated using accuracy, precision, recall, F1-score and ROC-AUC, rather than accuracy alone, because the high-demand class is smaller than the non-high-demand class.

Derived features we can create

Derived feature	Source	Why useful
title_length	name	Longer or more descriptive titles may affect demand
title_word_count	name	Simple NLP signal
title keywords	name	Words like “private”, “cozy”, “central”, “subway” may signal positioning
log_price	price	Reduces effect of extreme prices
log_minimum_nights	minimum_nights	Handles outliers such as very long minimum stays
availability_band	availability_365	Groups listings into low, medium, high availability
review_recency_days	last_review	Optional, but should be treated carefully
map patch image	latitude, longitude + NYC map	Allows CNN-based spatial analysis

Model Design and Architecture

We will test whether different deep learning representations (structured data, listing-title text, and map-based spatial context) improve the identification of high-demand Airbnb listings.

Branch 1: Structured features

numeric + categorical features → dense layers

- price
- minimum_nights
- availability_365
- latitude
- longitude
- room_type
- neighbourhood_group
- neighbourhood
- calculated_host_listings_count

Branch 2: Text features

listing name → tokenisation → embedding → pooling layer

- name/title text
- tokenised words or embeddings

Branch 3: CNN spatial image features (exploratory spatial-context experiment)

map patch → Conv2D → MaxPooling → Conv2D → Flatten

- small map crop around listing coordinate

Output (Combined)

concatenate all branches → dense layer → sigmoid output

High-demand listing: yes/no

Model	Purpose
Model 1: Tabular neural network	Baseline deep learning model
Model 2: Text-only neural network	Tests whether listing titles contain useful demand signals
Model 3: CNN map model	Tests whether map-based spatial context helps
Model 4: Hybrid model	Combines structured, text, and CNN features

Team Responsibilities & ePortfolio Artefacts

Person	Report responsibility	ML/ePortfolio artefact
1 - Ross	Business question, strategic value, and success metric	Demand-proxy validation notebook: create an exploratory high_demand proxy from reviews_per_month, test whether high-demand listings differ by borough, room type, price band, availability band, and title length, and explain whether the proxy creates business-plausible patterns
2 - Chantelle and Sonya	Dataset audit, cleaning, missing values, outlier handling, final modelling dataset, and structured/tabular neural network baseline	Preprocessing + tabular NN notebook: clean the dataset, handle missing values and outliers, create the final modelling dataset, encode categorical variables, scale numeric features, and train the structured neural network baseline
3 - Veli	Text-based deep learning from the name column	NLP/text model notebook: clean listing titles, tokenise text, derive title-length, keyword and optional title-sentiment features, then train a text-based neural network to test whether listing-title language helps predict high demand
4 - Luis	CNN map-based spatial analysis	CNN spatial notebook: convert latitude/longitude to approximate map pixels, extract map patches, show sample patches, and train a CNN branch to test whether map-derived spatial context helps predict high demand
5 - Chantelle and Sonya	Model comparison, visualisations, recommendations, and limitations	Evaluation and comparison notebook/dashboard: compare tabular, text, CNN, and hybrid models using confusion matrix, precision, recall, F1, ROC-AUC or PR-AUC, then translate results into business recommendations

Person 1 = validates whether the business question and demand proxy make sense

Person 2 = builds the clean dataset and tabular neural network baseline

Person 3 = builds the text/NLP neural network branch

Person 4 = builds the CNN map-patch branch

Person 5 = compares models and turns results into recommendations

1000-Word Report Structure

1. Title / Introduction: 120 words

Introduce the project aim, dataset, prediction task, and confirm that the report follows the Deep Learning track.

2. Business Question: 120 words

State the Airbnb demand-prediction question and explain why identifying high-demand listings has business value.

3. Methodology: ML Approach Based on Track, 280 words

- Explain the dataset, demand proxy, feature engineering, excluded leakage features,
- The deep learning model variants: tabular, text/NLP, CNN map-patch, and optional hybrid model.

4. Analysis & Results: 220 words

Compare model performance using key metrics such as precision, recall, F1-score, confusion matrix, ROC-AUC or PR-AUC.

5. Visualisations: 90 words

Present the main business-friendly charts, including demand by borough, room type, price band, title length, map visualisation, and model comparison.

6. Discussion: Insights for Airbnb, 120 words

Explain the main business insights, practical recommendations, and limitations, including review activity as a proxy and the exploratory CNN approach.

7. Conclusion: 50 words

Summarise the answer to the business question, the value of the model comparison, and the main limitation.

8. Referencing: Excluded from word count

Use Harvard referencing for the business context, demand proxy, deep learning methods, and limitations.

9. Appendices / E-Portfolio Evidence: Excluded from word count

Store notebooks, code, extra analysis, screenshots, model outputs, and supporting visualisations in the e-portfolio.

Key Risks / Limitations

- Demand is estimated from review activity, not actual booking data.
- The CNN map branch is exploratory because the map is not a rich property image source.
- The map-to-coordinate alignment may be approximate.
- The model can identify associations, but it cannot prove that specific features cause higher demand.